M13 Batch 4 — Full Set

M13.3 Autoflight (ATA 22) · PDF pages 232–321

Flight director and take-off mode, auto-throttle, yaw damping, all weather operation and the ILS categories, system redundancy, helicopter stability augmentation.

293 questions · Level 3: 90 · Level 2: 151 · Level 1: 52

● marks a question built on a fact your instructor highlighted or underlined in the manual (240 of 293). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Flight director, roll bar and pitch bar

PDF pages 232–238 · 30 questions

LEVEL 1 — RECALL

1. Which two tasks does the flight director accomplish ●
   a. Commands for manual control and automatic trim of the pitch axis
   b. Monitoring of autopilot activities and selection of the thrust rating
   c. Commands for manual control and monitoring of autopilot activities

2. During which phases is a yaw bar displayed and the roll bar removed ●
   a. Flare below 30 ft radio height, rollout and take-off
   b. Align below 30 ft radio height, rollout and approach
   c. Flare below 50 ft radio height, climb and take-off

3. Which bars display the attitude reference commands in an ADI or PFD ●
   a. A pitch bar, a roll bar and eventually a yaw bar
   b. A pitch bar, a track bar and eventually a roll bar
   c. A roll bar, a speed bar and eventually a yaw bar

4. Which three speeds do the pilots determine during flight preparation for the take off phase ●
   a. V1, V2 and VLS
   b. VR, V2 and VMAX
   c. V1, VR and V2

5. At which height above the runway does the take off safety speed have to be obtained with one engine inoperative ●
   a. 50 ft
   b. 30 ft
   c. 35 ft

LEVEL 2 — UNDERSTAND AND EXPLAIN

6. Why are the flight director indicators overlaid on the artificial horizon ●
   a. They display speed commands affecting the linear degrees of freedom
   b. They display attitude commands affecting the rotary degrees of freedom
   c. They display track commands affecting the lateral degrees of freedom

7. An aircraft with a crossbar flight director is following its defined flight path. Where do the two bars lie?
   a. They lie apart at an equal distance above the horizon line
   b. They intersect at the middle mark of the own aircraft symbol
   c. They rest at the upper edge of the own aircraft symbol

8. Which rotations does the yaw bar command ●
   a. Rotations around the vertical axis only
   b. Rotations around the longitudinal axis only
   c. Rotations around the lateral axis only

9. The yaw bar in the ADI deflects during the take off run. Which control inputs does the pilot make? ●
   a. Aileron or roll spoiler inputs
   b. Elevator or stabiliser trim inputs
   c. Rudder or nose wheel steering inputs

10. Below 50 ft radio altitude on an autoland the flare, align and rollout modes are selected automatically. What part does the flight director then play? ●
    a. It serves only for monitoring the autopilot
    b. It serves only for commanding manual control inputs
    c. It is removed from the display until roll out has ended

11. The autopilot is engaged and a lateral mode is chosen on the control panel. What happens to the flight director? ●
    a. The flight director reverts to its take off mode
    b. The selected mode also holds for the flight director
    c. The flight director bars are biased out of view

12. During a flight path mode the aircraft adopts the commanded attitude. What do the command bars do?
    a. They stay deflected until the mode is deselected
    b. They move to a steady offset from the centre
    c. They return immediately to their centre position

13. The pitch bar deflects upward. Which input does it call for? ●
    a. Push elevator, pitch down
    b. Pull elevator, pitch up
    c. Right aileron, turn right

14. Which input does a deflection of the roll bar to the right signify ●
    a. Right aileron, turn right
    b. Left aileron, turn left
    c. Right rudder, yaw right

15. The yaw bar deflects to the left. Which input does that command? ●
    a. Right rudder pedal, yaw right
    b. Left rudder pedal, yaw left
    c. Left aileron wheel, turn left

16. The roll bar in the ADI deflects. Which autopilot signal does that deflection correspond to? ●
    a. The attitude error computed in the roll outer control loop
    b. The signal fed to the roll actuator in the inner roll loop
    c. The signal fed to the roll actuator in the outer heading loop

17. Which action removes a roll attitude error that the flight director indicates

- **a.** A reset made in the outer heading loop
- **b.** A servo input made by the roll actuator
- **c.** A control input made by the pilot himself

18. Why must an aircraft be able to hold different steady pitch attitudes?

- **a.** Because they depend on runway heading, visibility and flap setting
- **b.** Because they depend on thrust rating, bleed air load and temperature
- **c.** Because they depend on airspeed, centre of gravity and gross weight

19. Why are distances and airspeeds laid down in the airworthiness standards ●

- **a.** To guarantee a sufficient safety margin in all flight phases
- **b.** To allow a take off at the lowest possible runway visibility
- **c.** To keep the engine ratings within their derated temperature limits

20. Why does an aircraft with a tricycle gear have to be rotated at VR ●

- **a.** It accelerates on its nose gear and has to unload that gear
- **b.** It accelerates at a low angle of attack and needs more lift
- **c.** It accelerates at a high angle of attack and needs less drag

LEVEL 3 — APPLY AND REASON

21. A flight director shows a roll attitude error and the pilot moves the wheel to follow it. Which part of the loop structure has the pilot just closed?

- **a.** The attitude limiter path
- **b.** The outer heading loop
- **c.** The inner control loop

22. Why does integration have to be avoided when a vertical mode drives the flight director pitch bar

- **a.** The outer loop error is already proportional to attitude
- **b.** The behaviour of the pilot cannot be predicted
- **c.** The pitch bar signal is inverted before it is displayed

23. What would a delayed pilot reaction do to the integrator in the outer loop

- **a.** Drive it towards zero and centre the pitch bar
- **b.** Drive it towards saturation and wind up the pitch bar
- **c.** Hold it at a constant value and freeze the pitch bar

24. Why can the outer loop error not be used directly as the pitch attitude reference for the inner loop

- **a.** A zero error would then demand a pitch attitude of 20 degrees
- **b.** A zero error would then demand a constant pitch rate
- **c.** A zero error would then demand a pitch attitude of 0 degrees

25. The outer loop error is zero and the aircraft is held at a pitch attitude inside the suppression range. Where does the flight director pitch bar sit? ●

- **a.** Deflected fully up
- **b.** In its neutral position
- **c.** Deflected fully down

26. The aircraft pitches up beyond the suppression range. What is added to the pitch bar deflection? ●

- **a.** A command proportional only to the outer loop error
- **b.** A fixed command until the attitude re-enters the range
- **c.** An opposing command proportional to the exceeding angle

27. The pilot makes no input in answer to a short-term pitch bar deflection. What happens to the bar?

- **a.** It stays deflected until the mode is reselected
- **b.** It returns to its initial position again
- **c.** It increases its deflection steadily

28. The pilot follows a short-term pitch bar deflection caused by a gust inside the suppression range. What keeps that deflection from winding up?

- **a.** The washout in the pitch rate feedback path
- **b.** The attitude limiter at the pitch bar output
- **c.** The integrator in the outer control loop

29. An engine fails while the aircraft is accelerating and is still below V1. What must be done? ●

- **a.** The aircraft has to be rotated at once
- **b.** The take off has to be discontinued
- **c.** The take off has to be continued

30. An aircraft is in a V2 climb after an engine failure and below the acceleration altitude. What is the state of the gear and the flaps? ●

- **a.** Gear retracted, take off flaps already retracted
- **b.** Gear extended, take off flaps already retracted
- **c.** Gear retracted, take off flaps still extended

Answers — Flight director, roll bar and pitch bar

1. **c** The flight director has exactly two tasks: it gives commands for manual control, and it allows monitoring of the autopilot. (L1, PDF p. 232)

2. **a** Those are the three phases printed for yaw bar display. (L1, PDF p. 232)

3. **a** Pitch and roll bars, and possibly a yaw bar, carry the attitude reference commands. (L1, PDF p. 233)

4. **c** The three take off speeds named are V1, VR and V2. (L1, PDF p. 238)

5. **c** V2 is defined at 35 ft above the runway surface. (L1, PDF p. 238)

6. **b** Because the commands concern the rotary degrees of freedom they belong on the artificial horizon. (L2, PDF p. 232)

7. **b** On path the roll bar crosses the pitch bar at the middle mark of the aircraft symbol. (L2, PDF p. 232)

8. **a** The yaw bar is restricted to rotations about the vertical axis. (L2, PDF p. 232)

9. **c** Yaw bar commands are flown with rudder or nose wheel steering. (L2, PDF p. 232)

10. **a** In the autoland modes the flight director is a monitoring display only. (L2, PDF p. 232)

11. **b** A mode selected with the autopilot engaged applies to the flight director as well. (L2, PDF p. 232)

12. **c** The bars show attitude error only, so they centre as soon as the error is gone. (L2, PDF p. 233)

13. **b** An upward pitch bar asks for elevator back pressure to pitch up. (L2, PDF p. 233)

14. **a** A roll bar to the right asks for right aileron and a turn to the right. (L2, PDF p. 233)

15. **b** A yaw bar to the left asks for left rudder pedal to yaw left. (L2, PDF p. 233)

16. **b** The bar is driven by the same signal the autopilot would send to its roll actuator in the inner loop. (L2, PDF p. 234)

17. **c** The flight director only displays the command; the correction comes from the pilot's own input. (L2, PDF p. 234)

18. **c** Airspeed, CG position and gross weight each change the steady attitude required. (L2, PDF p. 236)

19. **a** They exist to give a sufficient safety margin in every flight phase. (L2, PDF p. 238)

20. **b** The ground run is at a low angle of attack, so rotation is needed to make enough lift. (L2, PDF p. 238)

21. **c** In flight director operation the pilot himself closes the inner loop that the autopilot servo would otherwise close. (L3, PDF p. 234)

22. **b** A pilot may react late or not at all, so an integrator cannot be used. (L3, PDF p. 236)

23. **b** The integrator saturates, the bar winds up and the command becomes useless. (L3, PDF p. 236)

24. **c** With the outer loop error as reference, zero error would force the attitude to zero as well. (L3, PDF p. 236)
25. **b** With no outer loop error and the attitude still inside the suppression range, the feedback of actual pitch attitude is suppressed, so the bar rests in its neutral position. (L3, PDF p. 236)
26. **c** Outside the range an opposing command sized by the exceeding angle is added to the bar. (L3, PDF p. 236)
27. **b** The washout lets the deflection decay back to the initial position. (L3, PDF p. 236)
28. **a** The washout makes the pitch rate feedback decay, so no lasting build-up occurs. (L3, PDF p. 236)
29. **b** Before V1 an engine failure obliges the crew to stop the take off. (L3, PDF p. 238)
30. **c** The gear comes up but the take off flaps stay out until the acceleration altitude. (L3, PDF p. 238)

Take-off mode

PDF pages 240–246 · 29 questions

LEVEL 1 — RECALL

31. For which system is the take off mode exclusively reserved ●

a. The flight director
b. The autopilot
c. The auto-throttle

32. The take off mode has just been selected on the ground. Which flight director channels are activated? ●

a. The pitch channel and the roll channel
b. The roll channel and the yaw channel
c. The pitch channel and the yaw channel

33. At which typical airspeed is the rotation attitude indicated on the pitch bar during the acceleration ●

a. Typically 80 kts
b. Typically 60 kts
c. Typically 100 kts

34. From which parameters is the potential flight path angle calculated ●

a. Longitudinal and vertical acceleration and angle of attack
b. Vertical acceleration, airspeed and angle of attack
c. Longitudinal acceleration, pitch rate and radio altitude

35. How is the actual flight path angle obtained ●

a. From the difference between pitch attitude and the threshold angle
b. From the difference between pitch attitude and angle of attack
c. From the sum of the pitch attitude and the angle of attack

LEVEL 2 — UNDERSTAND AND EXPLAIN

36. In which mode does an armed auto-throttle system stay throughout the take off phase ●

a. Mach mode
b. Thrust mode
c. Speed mode

37. A take off is to be made at a visibility between 125 m and 75 m. Which indication gives the pilot his lateral guidance? ●

a. A roll bar controlled by the runway heading signal
b. A pitch bar controlled by the radio altitude signal
c. A yaw bar controlled by the localizer deviation signal

38. Why are supplemental display systems fitted for low visibility take offs ●

a. They replace the artificial horizon during the take off run
b. They avoid a change between head-up and head-down views
c. They allow the roll bar to be shown below 30 ft radio height

39. How is the flight director switched into take off mode ●

a. Pressing a TOGA switch or pushing the levers into the TOGA detent
b. Arming the auto-thrust and selecting the departure runway by hand
c. Pressing a NAV switch or pushing the levers into the idle detent

40. The aircraft passes 30 ft radio height with no FMS flight plan active. Which lateral guidance does the roll bar give? ●

- **a.** It maintains the runway centre line
- **b.** It maintains the heading entered on the control panel
- **c.** It maintains the runway track

41. When is the pitch attitude to be adopted at VR shown on the pitch bar ●

- **a.** As soon as the gear struts extend, or at 35 ft above the runway
- **b.** As soon as the take off mode is selected, or at a certain airspeed
- **c.** As soon as the thrust levers reach the TOGA detent, or at lift off

42. Which airspeed does the flight director command for the initial climb at a normal take off ●

- **a.** An actual airspeed 10 to 20 kts above V2
- **b.** An actual airspeed equal to V2
- **c.** An actual airspeed 10 to 20 kts above VR

43. A take off run is in progress. Which reference does the pilot use to find the point of rotation?

- **a.** Airspeed only
- **b.** Radio height only
- **c.** Pitch bar position only

44. Which two inputs does the flight director need for yaw bar control up to 30 ft radio height ●

- **a.** Runway heading and localizer deviation
- **b.** Runway heading and radio altitude
- **c.** Localizer deviation and ground speed

45. Why does pitch attitude replace angle of attack in the calculation of the potential flight path angle during the take off phase ●

- **a.** Because the inertial system is not yet fully aligned
- **b.** Because the angle of attack vanes are not yet heated
- **c.** Because of the air flow conditions at that time

46. To which autopilot mode is the flight director take off yaw control mode equivalent ●

- **a.** The autoland roll-out mode
- **b.** The autoland align mode
- **c.** The autoland flare mode

47. Which input tells the yaw controller the direction in which the aircraft should be pointing? ●

- **a.** The localizer deviation signal
- **b.** The supplied runway heading
- **c.** The radio height signal

48. Which input lets the yaw controller recognise a departure from the runway centre line? ●

- **a.** The nose wheel steering position
- **b.** The localizer deviation signal
- **c.** The supplied runway heading

LEVEL 3 — APPLY AND REASON

49. The aircraft climbs through a radio altitude of 30 ft after lift off. What changes in the flight director? ●

- **a.** The yaw channel is deactivated and the roll channel activated
- **b.** The roll channel is deactivated and the yaw channel activated
- **c.** The pitch channel is deactivated and the roll channel activated

50. Which guidance does the yaw bar give from brake release up to a radio height of 30 ft ●

- **a.** A return to the runway centre line at any deviation
- **b.** A return to the runway track after a lateral mode change
- **c.** A turn onto the standard instrument departure

51. An FMS flight plan is active as the aircraft climbs through 30 ft radio height. What does the lateral command follow? ●

- **a.** The runway centre line until 1500 ft
- **b.** The localizer of the departure runway
- **c.** The standard instrument departure

52. An engine fails after V1. Which pitch attitude does the flight director command? ●

- **a.** One that reaches the safety speed plus 15 kts and then holds it
- **b.** One that reaches the safety speed at 35 ft and then holds it
- **c.** One that holds the rotation attitude until the acceleration altitude

53. The pitch bar already shows a rotation attitude while the aircraft is still accelerating below VR. What must the pilot do?

- **a.** Disregard that pitch bar command
- **b.** Rotate the aircraft into the pitch bar
- **c.** Reduce thrust until the rotation speed is reached

54. The take off runway has been selected at flight preparation. How does the yaw channel obtain its two inputs?

- **a.** From the IRS, which supplies the track and the drift angle
- **b.** From the FMS, which tunes the ILS and supplies the heading
- **c.** From the ADC, which supplies the heading and the deviation

55. Heading mode is selected for lateral guidance after the yaw bar has been removed. Which heading is followed? ●

- **a.** The heading at its activation or one entered on the panel
- **b.** The heading of the standard instrument departure leg
- **c.** The runway heading supplied by the navigation database

56. During take off the potential flight path angle drops below the threshold while the speed is already above V2. What does the autoflight system do? ●

- **a.** It raises the speed target by a further 15 kts
- **b.** It reduces the speed target to the safety speed
- **c.** It tries to maintain the speed held at engine failure

57. The potential flight path angle falls below the threshold while the aircraft is still below the safety speed. What becomes the speed target? ●

- **a.** VLS
- **b.** VR
- **c.** V2

58. Which difference remains between the flight director take off yaw control mode and the equivalent autopilot mode ●

- **a.** The localizer deviation signal is not made use of
- **b.** The pilot is actively involved in the loop
- **c.** The rudder is driven directly by the yaw servo

59. The pilot flies the yaw bar commands accurately during the take off run. What happens to the localizer deviation and the heading error? ●

- **a.** Both are frozen at their present values
- **b.** Both are replaced by a roll demand
- **c.** Both of them are progressively minimized

Answers — Take-off mode

31. **a** The take off mode belongs to the flight director alone. (L1, PDF p. 240)

32. **c** Selection of the take off mode brings up the pitch and yaw channels. (L1, PDF p. 240)

33. **b** The printed typical value is 60 kts. (L1, PDF p. 242)

34. **a** Those three parameters are the printed inputs to the potential flight path angle. (L1, PDF p. 244)

35. **b** It is the pitch attitude angle less the angle of attack. (L1, PDF p. 244)

36. **b** Through the whole take off the armed system remains in thrust mode. (L2, PDF p. 240)

37. **c** In that visibility band the yaw bar driven by localizer deviation provides the lateral guidance. (L2, PDF p. 240)

38. **b** They let the pilot see the guidance while still looking at the runway lighting. (L2, PDF p. 240)

39. **a** Either the TOGA switch or the TOGA detent selects the mode. (L2, PDF p. 242)

40. **c** Above 30 ft the roll bar holds the runway track until another lateral mode is chosen. (L2, PDF p. 242)

41. **b** Two implementations exist: at mode selection, or on passing a certain airspeed. (L2, PDF p. 242)

42. **a** On all engines the initial climb is flown 10 to 20 kts above V2. (L2, PDF p. 242)

43. **a** The rotation point is set by airspeed alone, read from the ASI. (L2, PDF p. 242)

44. **a** Yaw bar control needs the runway heading and the localizer deviation. (L2, PDF p. 242)

45. **c** The air flow conditions at take off make the angle of attack unusable, so pitch attitude is taken. (L2, PDF p. 244)

46. **a** It is the same control arrangement as the autoland roll-out mode. (L2, PDF p. 246)

47. **b** Alignment with the runway comes from the supplied runway heading. (L2, PDF p. 246)

48. **b** Deviations from the centre line are seen in the localizer deviation signal. (L2, PDF p. 246)

49. **a** At 30 ft the yaw channel drops out and the roll channel comes in at the same moment. (L3, PDF p. 240)

50. **a** Up to 30 ft the yaw bar works the aircraft back onto the runway centre line. (L3, PDF p. 242)

51. **c** With an active flight plan the lateral command follows the SID instead of the runway track. (L3, PDF p. 242)

52. **b** After an engine failure the commanded attitude reaches V2 at 35 ft and keeps it through the first and second segment climb. (L3, PDF p. 242)

53. **a** The flight director cannot find the rotation point, so a command before VR is ignored. (L3, PDF p. 242)

54. **b** The FMS tunes the ILS for the localizer deviation and reads the runway heading from its database. (L3, PDF p. 242)

55. **a** Heading mode follows either the heading at activation or the heading set on the control panel. (L3, PDF p. 242)

56. **c** Above V2 the system holds the speed the aircraft had at the failure instead of reducing the target. (L3, PDF p. 244)

57. **c** The target is automatically reduced to V2. (L3, PDF p. 244)

58. **b** The one difference is that the pilot sits in the loop. (L3, PDF p. 246)

59. **c** Following the bar drives both the lateral deviation and the heading error towards zero. (L3, PDF p. 246)

Auto-throttle: thrust, speed and Mach modes

PDF pages 248–262 · 59 questions

LEVEL 1 — RECALL

60. Which computer function determines the thrust ratings ●

a. The envelope protection function of the flight control computers
b. The thrust limit function of the flight management computers
c. The power management function of the engine computers

61. In which two areas can an auto-throttle or auto-thrust system relieve the pilot ●

a. Thrust control or pitch trim control
b. Thrust control or airspeed control
c. Airspeed control or lateral guidance

62. Which two modes of operation does an auto-throttle or auto-thrust system provide ●

a. Thrust mode and alpha floor mode
b. Speed or Mach mode and retard mode
c. Thrust mode and speed or Mach mode

63. For how long is the maximum take off thrust rating basically available ●

a. 15 minutes
b. 5 minutes
c. 2 minutes

64. Which rating is the highest thrust available without a time limit ●

a. Maximum climb thrust
b. Maximum take off thrust
c. Maximum continuous thrust

65. The auto-throttle system works in thrust mode provided the autopilot vertical mode is of which kind ●

a. A speed or Mach mode
b. An altitude hold mode
c. A vertical speed mode

66. The rearward throttle movement to the idle position during cruise-descent and flare carries which name ●

a. Retard bias
b. Rate feedback
c. Torque limiter

67. In an auto-thrust system the required rating mode is brought in by which action ●

a. The flight management system commanding the levers
b. A servo motor driving the thrust levers forward
c. The pilot setting the thrust levers by hand

68. In speed or Mach mode the auto-throttle system holds which quantity ●

a. A pre-selected airspeed or Mach number
b. A pre-selected rating for the flight phase
c. A pre-selected Mach trim setting

LEVEL 2 — UNDERSTAND AND EXPLAIN

69. What does the auto-thrust system do in place of moving the throttles ●

a. It controls the thrust over the servo clutch
b. It controls the thrust over a data bus
c. It controls the throttle cables over the FADEC

70. An aircraft has an auto-thrust system fitted. How can its throttles be moved? ●

a. Only manually
b. Manually or by the servomotor
c. Only automatically

71. What does the position of the throttles determine on an aircraft with an auto-thrust system ●

a. The reference speed used by the speed mode of the system
b. The thrust rating and so the limits for that flight phase
c. The thrust demand sent straight to the fuel control unit

72. What supported the development of the auto-thrust system ●

a. The derating of the turbine inlet temperature limits
b. The electronic control of modern turbofan engines
c. The introduction of the para visual display systems

73. Which demands was the auto-thrust system introduced to meet

a. Higher thrust ratings and a shorter take off run
b. Lower fuel burn and a wider speed envelope
c. Lower complexity and a better man/machine interface

74. Which settings can be made in thrust mode ●

a. Reference speed or Mach number
b. Thrust ratings or thrust idle
c. Thrust ratings or alpha floor

75. What is obtained when thrust idle is set ●

a. The lowest rating allowed for that flight phase
b. Only residual thrust from an idling engine
c. A thrust that just balances the aerodynamic drag

76. The system is working in speed or Mach mode. Where does its reference speed come from? ●

a. The autoflight control panel or the EEC
b. The autoflight control panel or the FMS
c. The air data computer or the FMS

77. From which low speed value do the protective functions keep the aircraft away ●

a. VLS
b. VR
c. V1

78. With respect to what does the overspeed protection work ●

a. Airspeed and the thrust rating set for that flight phase
b. Angle of attack and the pitch attitude during the approach
c. Altitude and the slat/flap or landing gear configuration

79. In which mode does the auto-thrust system observe the thrust ratings as a protective function ●

a. The speed or Mach mode
b. The idle thrust mode
c. The take off thrust mode

80. Which rule governs the control of airspeed by an autoflight system ●

a. Only one system can control it exclusively at any moment
b. Both systems control it together at any moment
c. The system with the smaller error controls it at any moment

81. Which system holds priority for airspeed control ●

a. The flight management system computer
b. The autopilot or flight director pitch channel
c. The auto-throttle or auto-thrust system

82. Why is the maximum power available from a turbofan engine reduced in service

a. To keep the thrust within the capacity of the data bus
b. To reach acceptable life cycles for highly stressed parts
c. To allow the thrust to be measured directly in flight

83. Which parameter limits the thermodynamic power of a turbofan engine ●

a. The maximum engine pressure ratio
b. The maximum low pressure rotor speed
c. The maximum turbine inlet temperature

84. The auto-throttle is working in thrust mode. Why are N1 or EPR used as the reference values? ●

a. Thrust is a force and varies with the bleed air load in flight
b. Thrust is a force and cannot be measured directly in flight
c. Thrust is a force and is limited by the turbine inlet temperature

85. Which period do most authorities apart from the FAA accept for engine failure performance calculations

a. 10 minutes of maximum take off thrust
b. 15 minutes of maximum continuous thrust
c. 5 minutes of maximum take off thrust

86. Conditions allow a reduced thrust take off. Why is that reduction recommended? ●

a. It prolongs engine life and lowers the failure probability
b. It shortens the take off run and lowers the noise footprint
c. It raises the thrust rating left available for the climb phase

87. How is go around thrust defined ●

a. The max take off rating corrected for the higher airspeed
b. The max continuous rating corrected for the higher altitude
c. The max climb rating corrected for the lower air density

88. Idle thrust has to be set with care on every engine type, so what must it still guarantee? ●

a. Bleed air delivery at the regulated cabin pressure
b. Spool-up to go-around power in the regulated time
c. Minimum fuel flow throughout the whole descent

89. Apart from other factors, idle thrust depends on which characteristic of the engine fitted ●

- **a.** Its acceleration characteristic
- **b.** Its bleed extraction characteristic
- **c.** Its compression characteristic

90. The permanent rating calculation takes account of which two inputs

- **a.** Thrust lever angle and fuel flow
- **b.** Air data and bleed air demand
- **c.** Radio height and flap position

91. The crew has made no manual selection of the phase used for the rating — what else can supply it? ●

- **a.** The flight management system
- **b.** The autothrottle computer unit
- **c.** The engine control computer

92. Level change is engaged and the aircraft is descending — what thrust does the auto-throttle set? ●

- **a.** The take-off rating
- **b.** The climb rating
- **c.** Idle

93. Thrust mode is engaged on the auto-throttle — what type of control does the system itself apply? ●

- **a.** Open loop control
- **b.** Cascaded loop control
- **c.** Closed loop control

94. Thrust mode has just been selected — what does the determined thrust rating then serve as? ●

- **a.** The reference for N2 closed loop control by the autothrottle computer
- **b.** The reference for fuel flow control by the hydromechanical unit
- **c.** The reference for N1 closed loop control by the engine computer

95. The power management function works out the rating for which quantity ●

- **a.** Each engine bleed demand
- **b.** Each defined flight phase
- **c.** Each throttle position

96. In an auto-thrust system which device turns lever position into the signal sent to the engine control unit ●

- **a.** An angle transducer at the levers
- **b.** A torque limiter in the servo drive
- **c.** A rate gyro at the throttle quadrant

97. A technician finds no throttle servo motor on an auto-thrust aircraft — how is that to be judged? ●

- **a.** As a defect, since the servo must drive the levers to the rating
- **b.** As a defect, since without it the retard bias cannot operate
- **c.** As normal, since the levers of such a system are not driven automatically

98. The speed and Mach control structure is built on how many control loops

- **a.** A single loop
- **b.** Three parallel loops
- **c.** Two cascaded loops

99. The time response of the speed and Mach controller is fixed by which two factors ●

a. Air data computer lag and thrust lever angle range
b. Thrust change dynamics and aircraft aerodynamics
c. Servo rate feedback and clutch torque limiter setting

100. The engine control input lever position is fed back to the servo — which effect is this meant to cut down? ●

a. Overshoot from the longitudinal acceleration
b. Hysteresis from the mechanical components
c. Drift from the air data computer

LEVEL 3 — APPLY AND REASON

101. Which components of the earlier auto-throttle installation become superfluous with auto-thrust ●

a. Thrust rating computers, data buses and engine computers
b. Throttle levers, thrust limit displays and mode panels
c. Servomotors, clutch mechanisms, cables and TOGA switches

102. Why does the relief given by an auto-thrust system only materialise when the pitch axis is also under autopilot control? ●

a. Thrust changes trim the aircraft out and automatic trim is needed
b. The autopilot has to hold the attitude while the levers are moving
c. The pitch channel supplies the reference speed to the thrust computer

103. During an approach the angle of attack becomes extreme. Which protection acts and with which thrust? ●

a. Alpha floor protection, with go around thrust
b. Low speed protection, with climb thrust
c. Overspeed protection, with idle thrust

104. The pitch channel is given the task of holding a certain altitude. Which system then controls the airspeed? ●

a. The auto-throttle or auto-thrust system
b. The flight management system computer
c. The pitch channel, which keeps its priority

105. The pitch channel is the system controlling the airspeed. In which mode is the auto-throttle then working? ●

a. Speed mode, setting the reference airspeed
b. Thrust mode, setting a rating or idle thrust
c. Mach mode, setting the cruise Mach number

106. An aircraft takes off at a low mass from a long runway at a low air temperature. Which rating is recommended? ●

a. Maximum take off thrust
b. Maximum continuous thrust
c. Flexible take off thrust

107. A go around is flown at go around thrust. For how long may that thrust be held? ●

a. For the same period as maximum take off thrust
b. For as long as the approach has to be repeated
c. Until climb thrust is set at the acceleration altitude

108. The maximum take off thrust period has elapsed but maximum thrust is still needed. Which rating is used?

a. Maximum continuous thrust
b. Maximum climb thrust
c. Flexible take off thrust

109. The servo drive is running the throttles forward in thrust mode — when does the movement stop?

a. The engine reaches the commanded N1 at its control unit
b. The power management value equals the rating reference
c. The thrust lever angle transducer reaches its forward stop

110. The aircraft climbs in thrust mode while pressure and temperature change — why do the throttles stay put?

a. Both functions work from the same input data
b. The retard bias holds the levers against the idle stop
c. The servo drive is declutched above the transition altitude

111. Two functions both work out ratings in thrust mode — what separates one from the other? ●

a. One works only in flight, the other works only on the ground
b. One uses the flight phase, the other the lever position
c. One uses the air data, the other uses the bleed demand

112. An integrating element is used in the speed and Mach mode in order to obtain which result ●

a. Faster throttle travel at large speed errors
b. Less hysteresis in the linkage
c. A speed error of zero at the end

113. No hysteresis limit applies and the speed error grows — what happens to the rate of throttle movement?

a. It falls as the limiter takes over
b. It stays fixed until the error has gone
c. It rises in proportion to the error

114. Adding the engine control input position feedback to the servo makes which further item necessary ●

a. A torque limiter behind the servo clutch
b. An electronic integrator ahead of the servo
c. A rate gyro ahead of the band pass filter unit

115. The compromise between precise speed control and smooth thrust is reached by feeding back which signal

a. Actual throttle lever position
b. Longitudinal acceleration
c. Actual pitch attitude angle

116. A higher reference has been commanded and the aircraft is still accelerating with the error already gone — what stops an overshoot?

a. The retard bias driving the throttles to idle
b. The limiter holding the error at ten knots
c. The acceleration bias on the speed error

117. A head-on gust raises the indicated speed while the aircraft itself is slowing — what keeps the throttles still?

- **a.** The integrator inside the speed controller
- **b.** The acceleration bias on the airspeed error
- **c.** The hysteresis of the throttle control cables

118. A gust suddenly raises the airspeed and no acceleration feedback is fitted — what would the throttles do?

- **a.** No movement of the throttles
- **b.** A cut back of the throttles
- **c.** An advance of the throttles

Answers — Auto-throttle: thrust, speed and Mach modes

60. **c** Ratings come from the power management function in the EEC or ECU. (L1, PDF p. 248)

61. **b** The two areas of relief are thrust control and airspeed control. (L1, PDF p. 248)

62. **c** The system offers a thrust mode and a speed or Mach mode. (L1, PDF p. 250)

63. **b** The printed basic limit is 5 minutes. (L1, PDF p. 254)

64. **c** MCT is the highest thrust with no time limit attached. (L1, PDF p. 254)

65. **a** Thrust mode is entered only while the pitch channel is holding speed or Mach. (L1, PDF p. 256)

66. **a** The drive back to idle in those two phases is called the retard bias. (L1, PDF p. 256)

67. **c** The pilot picks the rating mode by hand on the levers. (L1, PDF p. 258)

68. **a** The mode holds the speed or Mach value selected by the pilot or by the FMS. (L1, PDF p. 260)

69. **b** Thrust is commanded over a data bus rather than by moving the levers. (L2, PDF p. 248)

70. **a** With auto-thrust the levers move by hand only. (L2, PDF p. 248)

71. **b** Lever position picks the rating, which sets the thrust limits for the phase. (L2, PDF p. 248)

72. **b** FADEC engine control made the auto-thrust system possible, and the functions are often built into it. (L2, PDF p. 248)

73. **c** It answered the demand for less complexity and a better man/machine interface. (L2, PDF p. 248)

74. **b** Thrust mode sets either a rating, that is a thrust limit, or thrust idle. (L2, PDF p. 250)

75. **b** Thrust idle leaves only the residual thrust of the idling engine. (L2, PDF p. 250)

76. **b** The reference is either set on the control panel or supplied by the FMS. (L2, PDF p. 250)

77. **a** VLS is the low speed limit the system protects. (L2, PDF p. 250)

78. **c** Overspeed protection takes account of altitude and the slat/flap or gear configuration. (L2, PDF p. 250)

79. **a** This protective function belongs to the SPD/Mach mode: the thrust ratings go on being observed while the system is controlling speed. (L2, PDF p. 250)

80. **a** Airspeed is controlled by one system at a time, never two. (L2, PDF p. 252)

81. **b** Priority lies with the pitch channel of the autopilot or flight director. (L2, PDF p. 252)

82. **b** Continuous maximum thermodynamic power would wear the engine out quickly. (L2, PDF p. 252)

83. **c** Turbine inlet temperature sets the thermodynamic ceiling, and is then derated. (L2, PDF p. 252)

84. **b** Thrust itself cannot be measured in flight, so derived parameters stand in for it. (L2, PDF p. 252)

85. **a** Those authorities allow 10 minutes of max take off thrust for that calculation. (L2, PDF p. 254)

86. **a** Reduced thrust extends engine life and cuts the chance of a failure. (L2, PDF p. 254)

87. **a** It is the max take off rating adjusted for the greater airspeed in a go around. (L2, PDF p. 254)

88. **b** Idle may not be so low that the engine cannot spool up to go-around thrust in the required time. (L2, PDF p. 256)

89. **a** How fast the engine accelerates decides how low idle may be set. (L2, PDF p. 256)

90. **b** Those are the two inputs named for the rating calculation. (L2, PDF p. 256)

91. **a** The flight phase comes either from the pilot or from the FMS. (L2, PDF p. 256)

92. **c** In a level change descent the auto-throttle sets idle thrust. (L2, PDF p. 256)
93. **a** In thrust mode the auto-throttle itself works open loop. (L2, PDF p. 256)
94. **c** The rating becomes the N1 reference that the engine computer then closes the loop on. (L2, PDF p. 256)
95. **c** Power management maps throttle position to a rating, phase by phase is the other function. (L2, PDF p. 256)
96. **a** An angle transducer turns lever position into the TLA signal sent to the ECU. (L2, PDF p. 258)
97. **c** An auto-thrust system has no lever servo at all, so none is missing. (L2, PDF p. 258)
98. **a** Speed and Mach control uses one single loop. (L2, PDF p. 260)
99. **b** Those are the two things named as setting the controller response. (L2, PDF p. 260)
100. **b** The position feedback is there to reduce mechanical hysteresis. (L2, PDF p. 260)
101. **c** Those mechanical items and the TOGA switches drop out with auto-thrust. (L3, PDF p. 248)
102. **a** Every thrust change puts the aircraft out of trim, and the automatic trim comes from the autopilot pitch channel. (L3, PDF p. 248)
103. **a** Alpha floor protection answers an extreme angle of attack by applying go around thrust. (L3, PDF p. 250)
104. **a** The pitch channel delegates speed control once it is busy with another task. (L3, PDF p. 252)
105. **b** When the pitch channel has the speed, the thrust system sits in thrust mode on a rating or idle. (L3, PDF p. 252)
106. **c** Those conditions are exactly the case for the reduced, flexible take off rating. (L3, PDF p. 254)
107. **a** Go around thrust carries the same time limit as maximum take off thrust. (L3, PDF p. 254)
108. **a** That abnormal case is one of the two uses printed for maximum continuous thrust. (L3, PDF p. 254)
109. **b** The servo runs until the fed back power management value cancels the rating reference in the summing point. (L3, PDF p. 256)
110. **a** Both values move together, so their difference stays zero and no servo signal is produced. (L3, PDF p. 256)
111. **b** Rating calculation is keyed to the flight phase and power management to lever position. (L3, PDF p. 256)
112. **c** Only an I-controller drives the remaining error to zero. (L3, PDF p. 260)
113. **c** Rate feedback makes throttle speed directly proportional to the error. (L3, PDF p. 260)
114. **b** The feedback destroys the servo's own integrating action, so an electronic integrator must replace it. (L3, PDF p. 260)
115. **b** Longitudinal acceleration feedback is the means of reconciling the two demands. (L3, PDF p. 262)
116. **c** The acceleration bias pulls the throttles back before the speed is reached. (L3, PDF p. 262)
117. **b** The acceleration bias cancels the false speed rise so the levers do not move. (L3, PDF p. 262)
118. **b** The loop would read the gust as excess speed and close the throttles. (L3, PDF p. 262)

Mode overview and yaw damping

PDF pages 266–270 · 21 questions

LEVEL 1 — RECALL

119. On a jet aircraft the swept wing reduces which stability, compared with the lateral stability ●

a. The directional static stability margin
b. The vertical dynamic stability
c. The longitudinal dynamic stability

120. The oscillation caused by longitudinal dynamic instability takes place about which axes ●

a. The yaw and pitch axes
b. The yaw and roll axes
c. The pitch and roll axes

121. The angular velocity about the yaw axis is sensed by which device ●

a. A pitot probe
b. An accelerometer
c. A rate gyro

122. The band pass filter in the yaw damper suppresses which unwanted signals ●

a. Structural oscillations sensed by the rate gyro
b. Speed adaption signals from the air data computer
c. Rudder pedal movements made by the pilot

123. An all weather operation approval may be restricted in which two ways ●

a. In scope and in time
b. In crew and in aircraft
c. In altitude and in speed

LEVEL 2 — UNDERSTAND AND EXPLAIN

124. Insufficient longitudinal dynamic stability spoils which two things ●

a. Range of the aircraft and the life of the engines
b. Controllability of the aircraft and passenger comfort
c. Accuracy of the ILS and of the automatic landing system

125. A jet is rolling and slipping sideways in a repeating four second cycle — what is this motion called? ●

a. Spiral dive
b. Phugoid
c. Dutch roll

126. A damper is fitted to stop the roll and slip motion of a swept wing jet — which surface does it move? ●

a. The ailerons
b. The spoilers
c. The rudder

127. A yaw damper is required during which part of the flight ●

a. The enroute phase
b. The take-off roll
c. The flare

128. The yaw rate signal is adapted to airspeed in which manner ●

a. Directly proportional to the airspeed
b. Independently of the airspeed
c. Inversely proportional to airspeed

129. A constant yaw rate signal reaching the band pass filter is treated in which way ●

a. It is passed unchanged
b. It is damped out
c. It is amplified

130. The yaw damper actuator output is combined with which other demand at the mechanical summing point

a. The rudder pedal input
b. The band pass filter output
c. The speed adaption signal

131. The yaw damper is making continuous rudder corrections in the cruise — what do the rudder pedals do? ●

a. They move in step with the rudder
b. They are locked until the damper drops out
c. Nothing, they stay where they are

132. An operator intends to fly in all weather conditions — which flight rules must its approval take in? ●

a. Visual flight rules
b. Special visual flight rules
c. Instrument flight rules

133. An operator gains an all weather approval in its air operators certificate — what does that approval cover? ●

a. Each individual aircraft registration
b. The aircraft type or series of types
c. Each individual flight crew member

134. The all weather approval holds good for which runways ●

a. Those approved by the authority for that category
b. Only those listed in the operator's own route manual
c. Only those fitted with a roll out guidance system

135. Directional guidance for take-off in low visibility is covered by which subpart of the all weather operations requirements

a. Subpart 2
b. Subpart 4
c. Subpart 1

LEVEL 3 — APPLY AND REASON

136. Dutch roll grows into a problem at high altitude for which reason ●

a. The lift of the swept wing rises with altitude
b. The rudder travel available falls off with altitude
c. The damping aerodynamic forces fall with altitude

137. The yaw damper becomes unserviceable during the cruise — what must then be done with the aircraft? ●

- **a.** It has to slow to the minimum airspeed
- **b.** It has to descend to a lower altitude
- **c.** It has to climb to a higher altitude

138. The aircraft accelerates in the cruise — what happens to the yaw rate signal passed on to the actuator? ●

- **a.** It is held at its take-off value
- **b.** It is built up as the speed rises
- **c.** It is cut back as the speed rises

139. The pilot presses a rudder pedal while the yaw damper is working — how do the two demands reach the rudder?

- **a.** The damper demand overrides any pedal demand made
- **b.** The pedal demand switches the damper out of use
- **c.** They are added or subtracted at a summing point

Answers — Mode overview and yaw damping

119. c The swept wing costs longitudinal dynamic stability relative to the lateral. (L1, PDF p. 266)

120. b The motion is a combined yaw and roll oscillation. (L1, PDF p. 266)

121. c Yaw rate is picked up by a rate gyro. (L1, PDF p. 268)

122. a Airframe structural modes sensed by the gyro are filtered out. (L1, PDF p. 268)

123. a Scope and time are the two named restrictions. (L1, PDF p. 270)

124. b Both handling and passenger comfort suffer. (L2, PDF p. 266)

125. c The combined roll and sideslip oscillation is named Dutch roll. (L2, PDF p. 266)

126. c The yaw damper works through the rudder. (L2, PDF p. 266)

127. a It is the cruise, at altitude, where the damper is needed. (L2, PDF p. 266)

128. c The adaptation is inversely proportional to speed. (L2, PDF p. 268)

129. b A steady yaw rate is damped by the filter rather than fed through. (L2, PDF p. 268)

130. a Damper demand and pedal demand are summed before the rudder actuator. (L2, PDF p. 268)

131. c In series mode a damper input does not move the pedals. (L2, PDF p. 268)

132. c The approval must cover IFR flight before all weather operation can follow. (L2, PDF p. 270)

133. b The approval is granted against a type or series of types. (L2, PDF p. 270)

134. a Approval follows the authority's own runway classification for the category. (L2, PDF p. 270)

135. b Subpart 4 deals with low visibility take-off guidance. (L2, PDF p. 270)

136. c Thin air means weaker aerodynamic damping, so the oscillation is no longer suppressed naturally. (L3, PDF p. 266)

137. b Lower down the natural aerodynamic damping is strong enough. (L3, PDF p. 266)

138. c Inverse proportionality means the signal is reduced as speed increases. (L3, PDF p. 268)

139. c Both demands are summed, so neither cancels the other. (L3, PDF p. 268)

All weather operation: definitions and RVR

PDF pages 272–278 · 28 questions

LEVEL 1 — RECALL

140. ILS precision approaches are subdivided into which categories ●

a. CAT I to CAT III
b. CAT I to CAT VIII
c. CAT A to CAT C

141. A low visibility take-off is any take-off made below which runway visual range ●

a. 550 metres
b. 300 metres
c. 400 metres

142. The decision height is measured above which reference ●

a. The touch down zone
b. The mean sea level datum
c. The runway threshold

143. The alert height is defined by which party ●

a. The airport operator
b. The aircraft manufacturer
c. The national aviation authority

144. A system failure during an approach can be met by which three courses of action ●

a. Continue, revert or go around
b. Continue, descend or go straight in
c. Revert, divert or go straight in

LEVEL 2 — UNDERSTAND AND EXPLAIN

145. A CAT III b approach is being planned — what decision height condition applies? ●

a. Below 50 ft or no decision height
b. No decision height and no visibility limit
c. Below 100 ft but not below 50 ft

146. An airport has no runway visual range measuring equipment — which value may be converted instead? ●

a. The meteorological visibility
b. The cloud base height measured
c. The decision height selected

147. The aircraft reaches the decision height without the required visual reference — what must the crew do? ●

a. Revert to a lower category
b. Continue down to the alert height
c. Initiate a missed approach

148. The alert height is quoted in an aircraft's approach data — which approaches does it belong to? ●

a. CAT I approaches flown with the autopilot engaged
b. CAT III approaches with a fail-operational autoland
c. CAT II approaches with a fail-passive autoland fitted

149. On what does the value chosen for the alert height depend ●

a. The reported runway visual range of the day
b. The landing distance available on the runway
c. The failure probability of the autoland

150. Where is the pilot to whom the runway visual range refers assumed to be ●

a. At the holding point beside the runway
b. Over the threshold on the glide path
c. On the centre line of the runway

151. The minimum runway visual range required for a CAT III approach depends on which two things ●

a. The equipment carried and the autoland capability
b. The decision height and the alert height selected
c. The runway length and the approach lighting fitted

152. The crew needs to know the capability of their automatic landing system — where do they read it? ●

a. As an amber light on the ILS deviation alert
b. As an approach category on the flight annunciator
c. As a decision height index on the radio altimeter

153. The crew decides to continue the planned approach after a failure — how far down may they go? ●

a. Its decision height or minimum descent altitude
b. The alert height of the automatic landing system
c. The touch down zone marking on the runway

154. A failure leaves the aircraft able to fly a lesser approach — what does reverting mean here? ●

a. Changing to a category the reported weather would allow
b. Changing to a category the runway lighting supports
c. Changing to a category the remaining equipment allows

155. For what purpose is a go around flown after a system fault ●

a. To investigate the failure and plan another approach
b. To allow the ground equipment to be reset by control
c. To reduce the landing weight before a second attempt

156. Besides the type of failure, what else decides the action to be taken

a. The category of runway lighting then in use
b. The age of the autoflight equipment then installed
c. The present position of the aircraft on approach

LEVEL 3 — APPLY AND REASON

157. An approach is planned to a decision height of 150 ft with a reported visual range of 400 m — which category applies? ●

a. CAT I
b. CAT II
c. CAT III a

158. The reported runway visual range is 100 m and the decision height is 40 ft — which category covers it? ●

a. CAT III a
b. CAT III c
c. CAT III b

159. A failure appears in the redundant part of the autoland system above the alert height — what follows? ●

a. The failure is ignored and the approach continued
b. The approach is broken off and a go around flown
c. The approach reverts to the next higher category

160. A CAT III A approach with a decision height below 100 ft is flown with a fail-passive landing system — what is the minimum visual range?

a. 200 m
b. 75 m
c. 125 m

161. The automatic landing system is fail-passive and the decision height is below 50 ft — is such an approach permitted, and to what visual range?

a. Yes, down to 75 metres
b. Yes, down to 125 metres
c. No, it is not authorised at all

162. A CAT III B approach to a decision height below 50 ft with a fail-operational system and a fail-passive roll-out system needs which minimum visual range

a. 125 m
b. 200 m
c. 75 m

163. An approach with no decision height is flown to a runway visual range of 75 m. Which standard must the roll-out system hold

a. Fail-passive
b. Fail-safe
c. Fail-operational

164. Two aircraft have a fail-operational landing system, one with a roll-out system and one without, both at a decision height below 50 ft — how are the two treated?

a. Both may fly it to the same visual range
b. Only the one without roll-out may fly the approach
c. Only the one with roll-out may fly the approach

165. In which band must a failure happen for a reversion to another approach category to be considered ●

a. Above 1000 ft above the ground
b. At or below the alert height
c. Below 1000 ft above the ground

166. A fail-operational autoflight system fails below 1000 ft but above the alert height — what action follows? ●

a. A reversion to CAT I is flown
b. A go around is executed
c. The approach continues to decision height

167. After a failure the weather allows a higher minimum and enough autoflight capability remains — what may the crew then do? ●

a. Descend below the alert height
b. Continue down to the original minimum
c. Fly that approach category instead

Answers — All weather operation: definitions and RVR

140. a The subdivision runs from CAT I to CAT III. (L1, PDF p. 272)

141. c Below 400 m RVR the departure counts as a low visibility take-off. (L1, PDF p. 272)

142. c Decision height is referenced to the runway threshold. (L1, PDF p. 274)

143. b It is the manufacturer who sets the alert height. (L1, PDF p. 274)

144. a Those three are the courses of action named. (L1, PDF p. 278)

145. a CAT III b covers decision heights under 50 ft and the no decision height case. (L2, PDF p. 272)

146. a Met visibility can be converted where no RVR equipment exists. (L2, PDF p. 272)

147. c Without the visual reference at decision height a missed approach is compulsory. (L2, PDF p. 274)

148. b The term belongs to fail-operational CAT III approaches. (L2, PDF p. 274)

149. c The alert height depends on one thing only: the failure probability of the autoland system. (L2, PDF p. 274)

150. c The definition places the pilot on the runway centre line. (L2, PDF p. 274)

151. a Installed equipment and autoland capability together fix the minimum RVR. (L2, PDF p. 276)

152. b Capability shows up as the available approach category on the flight mode annunciator. (L2, PDF p. 276)

153. a Continue means fly the planned approach to its own minimum. (L2, PDF p. 278)

154. c Reversion is driven by what the surviving equipment can still do. (L2, PDF p. 278)

155. a The go around buys time to look at the failure and reconfigure. (L2, PDF p. 278)

156. c Where the aircraft is on the approach matters as much as what failed. (L2, PDF p. 278)

157. b A decision height between 200 ft and 100 ft with at least 300 m visual range is CAT II. (L3, PDF p. 272)

158. c A decision height below 50 ft with a visual range between 200 m and 75 m falls in CAT III b. (L3, PDF p. 272)

159. b Above the alert height such a failure forces a go around. (L3, PDF p. 274)

160. a Every CAT III A column in the table shows 200 m. (L3, PDF p. 277)

161. c In the fail-passive column of the table the row for a decision height below 50 ft reads 'Not authorized'. (L3, PDF p. 277)

162. a Under the fail-operational landing system, the column for a fail-passive roll-out system gives 125 m at a decision height below 50 ft. (L3, PDF p. 277)

163. c In the CAT III RVR table the no decision height row gives 75 m only in the fail-operational roll-out column; the fail-passive roll-out column of that row reads 'Not authorized'. (L3, PDF p. 277)

164. c Without a roll-out system that decision height is not authorised at all. (L3, PDF p. 277)

165. a The general rule puts the reversion decision above 1000 ft AGL. (L3, PDF p. 278)

166. b In that band any failure of a fail-operational system means a go around. (L3, PDF p. 278)

167. c Both conditions together allow the higher minimum approach to be flown. (L3, PDF p. 278)

Objectives, safety and CAT III requirements

PDF pages 280–286 · 30 questions

LEVEL 1 — RECALL

168. The objective of CAT II autoflight operation is to hold which level of safety ●

a. One equivalent to manual flight operation
b. One set by the aircraft manufacturer alone
c. One above that of manual flight operation

169. In a fail-safe system the output signal after a failure takes which state ●

a. The last valid state before failure
b. A defined and equally safe state
c. An arbitrary but safe state

170. In a fail-passive system the output signal once the system has failed takes which state ●

a. A well defined failure state
b. An arbitrary but safe state
c. A state within acceptable limits

LEVEL 2 — UNDERSTAND AND EXPLAIN

171. The CAT II safety level depends partly on what the crew can see — which outside aid is named?

a. Approach and runway lighting
b. Runway friction measurement
c. Ground radar surveillance cover

172. The aircraft reaches the decision height on a CAT II approach — what is the position regarding a manual landing? ●

a. It is allowed but is not compulsory
b. It is compulsory in every single case
c. It is forbidden in every case

173. CAT III autoflight operation holds a safety level equivalent to manual flight under which conditions ●

a. Conditions no worse than CAT II minima
b. The worst weather and visibility
c. Normal weather and visibility conditions

174. To what extent is a manual landing possible from a CAT III decision height ●

a. It is possible but is not compulsory
b. It is compulsory for the pilot flying
c. It is normally not possible at all

175. An aircraft is to be certified for CAT III operation — which equipment is compulsory? ●

a. A flight director approach coupler
b. A head-up ground roll guidance display
c. An automatic or hybrid landing system

176. How does the reliability demanded of CAT III systems compare with that demanded for CAT II? ●

a. It is very much the same
b. It is considerably higher
c. It is somewhat lower still

177. A designer cuts the number of components in a system — why does availability improve?

a. Fewer components leave fewer ways to fail
b. Fewer components leave a faster response time
c. Fewer components leave a lower system weight

178. What must redundant systems show for CAT III operation ●

a. Automatic reset after every detected failure
b. Identical hardware from a single manufacturer
c. Defined behaviour for any conceivable failure

179. To behave passively when a fault appears a system needs which provision during normal operation ●

a. Periodic self testing carried out at every engine start
b. Manual selection of the standby channel by the crew
c. Permanent monitoring with suppression of any failure

180. A system that stays operative after a failure is described by which term ●

a. Failure neutral
b. Failure suppressed
c. Failure tolerant

181. What does a drop out of the main system cause with passive redundancy ●

a. An automatic shut down of all parallel systems
b. A manual selection of the remaining system
c. An automatic change to the redundant system

182. In passive redundancy the system held in reserve carries which name ●

a. The active parallel system
b. The hot standby system
c. The monitored main system

183. A main channel drops out and a reserve takes over automatically — what can be said of the system function afterwards? ●

a. The complete functionality is restored
b. Only the monitored functions are restored
c. Only half the functionality is restored

184. For decision heights below 100 ft the two ILS receivers must feed the pilot stations in which way

a. Both pilot stations from the same receiver
b. Both receivers into one combined display
c. Each pilot station from its own receiver

185. A decision height of 50 ft or greater is planned — what standard must the automatic landing system hold? ●

a. Fail-operational
b. Fail-passive
c. Fail-safe

186. A decision height below 50 ft is planned — what standard must the automatic go-around system hold? ●

a. Fail-operational
b. Fail-safe
c. Fail-passive

187. For a no decision height approach the automatic ground roll control may hold which standards ●

a. Fail-operational or fail-safe
b. Fail-passive or fail-safe
c. Fail-operational or fail-passive

188. An approach with no decision height is to be flown — what provision is required for speed control? ●

a. Automatic throttle control
b. Head-up speed guidance
c. Manual throttle control only

LEVEL 3 — APPLY AND REASON

189. What is the crew able to judge at a CAT III A decision height ●

a. Whether touch down will fall in the touch down zone
b. Whether the autoland has dropped to fail-passive
c. Whether the roll out can be supervised visually throughout

190. A CAT III B minimum is set so as to secure which crew capability ●

a. Visual acquisition of the approach light system
b. Visual supervision of the roll out
c. Visual confirmation of the touch down point

191. Automatic guidance of the roll out until taxi speed is reached is demanded for which categories ●

a. CAT II and CAT III A
b. CAT III A and CAT III B
c. CAT III B and CAT III C

192. One of several parallel channels drops out and the output signal is unaffected — which redundancy is in use? ●

a. Active redundancy
b. Passive redundancy
c. Monitored redundancy

193. An operator wants to certify a CAT II aircraft with no automatic throttle system — what must be shown?

a. That the engines accelerate within the regulated time
b. That speed can be held by hand without undue workload
c. That the autopilot can hold speed within ten knots

194. The number of ILS receivers and radio altimeters may have to be raised for which reason

a. To give an alert height indication at both stations
b. To give fail-operational capability where required
c. To give a roll out guidance display at both stations

195. The decision height planned is below 50 ft — what standard must the automatic landing system then meet? ●

a. Fail-operational
b. Fail-safe
c. Fail-passive

196. Which wheel provision is required for an approach flown with no decision height ●

- **a.** An anti-skid braking system
- **b.** A carbon brake wear indicator
- **c.** An autobrake disarm warning

197. A fail-passive approach system without automatic landing is acceptable at a decision height of 50 ft or greater provided the aeroplane holds which property

- **a.** A high approach speed and a long wheel base
- **b.** A fail-operational go-around capability fitted
- **c.** A low approach speed and good manoeuvrability

Answers — Objectives, safety and CAT III requirements

168. a The aim is equivalence with manual operation, under worse weather. (L1, PDF p. 280)

169. c Fail-safe gives an arbitrary state that is nevertheless safe. (L1, PDF p. 283)

170. a Fail-passive gives a defined, known failure state. (L1, PDF p. 283)

171. a Outside visual cues from the approach and runway lighting are listed. (L2, PDF p. 280)

172. a CAT II minima permit a manual landing without requiring one. (L2, PDF p. 280)

173. b CAT III carries that equivalence into the worst conditions. (L2, PDF p. 280)

174. c Unlike the lower categories, CAT III leaves no room for a manual landing from the decision height. (L2, PDF p. 280)

175. c CAT III requires an automatic or hybrid landing system. (L2, PDF p. 280)

176. b CAT III calls for considerably higher reliability than CAT II. (L2, PDF p. 280)

177. a Less complexity means fewer ways for the system to fail. (L2, PDF p. 283)

178. c The behaviour at every conceivable failure must be defined. (L2, PDF p. 283)

179. c Continuous monitoring plus suppression is what makes passive behaviour possible. (L2, PDF p. 283)

180. c Such a system is called failure tolerant. (L2, PDF p. 283)

181. c The change over happens automatically. (L2, PDF p. 283)

182. b The reserve system is called the hot standby. (L2, PDF p. 283)

183. a The standby restores the full function after the change over. (L2, PDF p. 283)

184. c The two stations must be fed from separate receivers. (L2, PDF p. 286)

185. b At 50 ft and above a fail-passive landing system is enough. (L2, PDF p. 286)

186. c The go-around system is required to be fail-passive there. (L2, PDF p. 286)

187. c Either standard is accepted for the ground roll function. (L2, PDF p. 286)

188. a Automatic throttle control is on the no decision height equipment list. (L2, PDF p. 286)

189. a At CAT III A the crew can still see whether the aircraft will land in the zone. (L3, PDF p. 280)

190. b The CAT III B minimum exists so the roll out can be watched. (L3, PDF p. 280)

191. c Flare and roll out guidance to taxi speed is listed for CAT III B and C. (L3, PDF p. 280)

192. a Only active redundancy keeps the output going with no switch over. (L3, PDF p. 283)

193. b The way out is an in-flight demonstration of manual speed control. (L3, PDF p. 284)

194. b Extra sensors are what makes a fail-operational installation possible. (L3, PDF p. 286)

195. a Below 50 ft only a fail-operational landing or hybrid system will do. (L3, PDF p. 286)

196. a An anti-skid braking system is one of the items listed for no decision height. (L3, PDF p. 286)

197. c Those handling properties are what make the manual landing acceptable. (L3, PDF p. 286)

Fail safe and fail passive systems

PDF pages 288–294 · 27 questions

LEVEL 1 — RECALL

198. Which system is defined as one comprising an automatic and/or a hybrid landing system ●

a. the flight control system
b. the landing system
c. the guidance system

199. Which equipment does the term landing system cover ●

a. onboard equipment only
b. onboard and ground equipment
c. ground equipment only

200. Which failure of a control surface can occur on a fail safe flight control system ●

a. a slow runaway of the control surface
b. a hard over of the control surface
c. a complete loss of control surface authority

201. The autopilot runs a surface hard over while the aircraft is enroute. Within which time must the pilot be able to take over control? ●

a. 1 second
b. 3 seconds
c. 10 seconds

202. Which kind of redundancy does the parallel operation of two fail safe autopilots show ●

a. standby redundancy
b. active redundancy
c. passive redundancy

203. Which arrangement inside one and the same autopilot gives fail passive behaviour ●

a. two autopilots coupled rigidly together
b. two independent channels working in parallel
c. three actuators driven by a single channel

204. What does the autopilot do when the comparison of its two channels detects a failure ●

a. it disengages
b. it reverts to one channel
c. it limits its authority

LEVEL 2 — UNDERSTAND AND EXPLAIN

205. Which items does the landing system comprise in accordance with JAR-AWO 321 ●

a. all instruments, displays and ground transmitters
b. all sensors, instruments and power supplies
c. all sensors, antennas and ground stations

206. What are the differing architectures of the two major transport aircraft manufacturers meant to realize

a. a common standard of autopilot engagement
b. a lower weight of the autopilot installation
c. failure modes acceptable for CAT III operation

207. Two fail passive single systems are operated together in dual operation. Which behaviour does the resulting system show? ●

a. fail operational behaviour
b. fail safe behaviour
c. fail passive behaviour

208. Why is a hard over not suppressed by a fail safe flight control system ●

a. because it cannot recognize the failure
b. because its servos have no force limiter
c. because its actuators have no clutch

209. How is the effect of a failure limited on a fail safe system

a. by comparing the two autopilot channels
b. by voting between three actuator outputs
c. by reducing the authority of the servos

210. Under which condition is authority limitation allowed at all ●

a. only if a manual flight control system exists
b. only if the servos carry a torque limiter
c. only if a second autopilot is installed

211. Which effect does a failure of a fail passive flight control system have on the trim state and the flight path ●

a. both of them are changed considerably
b. the trim state changes but the flight path does not
c. neither of them is changed considerably

212. What can no longer be carried out once a fail passive flight control system has failed ●

a. an ILS approach
b. an automatic landing
c. a manual go around

213. The fail passive automatic flight control system fails during an approach. What must happen then? ●

a. the system continues the approach and landing by itself
b. the second autopilot takes over control and lands
c. the pilot has to take over the control of the aircraft

214. How is the independence of two autopilots working in dual operation achieved ●

a. by one common computer serving both of the channels
b. by separate supplies, sensors, computers and actuators
c. by a common power supply feeding two sets of sensors

215. Which causes make the rigidly coupled actuators of two autopilots work against each other

a. tolerances, pressure differences or poor adjustment
b. different reference values in the two sensor units
c. a delay between the two autopilot computers

216. What is done for safety reasons in the two channels of one fail passive autopilot ●

a. one of the two channels is left unpowered as a spare
b. different hardware and different programming languages
c. the same hardware and the same programs in both lanes

217. A dual channel autopilot has produced an output in each of its channels. What happens to the two signals before the actuator sees them? ●

a. they are added together and then released
b. they are compared for validity and then released
c. they are sent on separately to two actuators

218. By which three means is fail passive behaviour reached in a dual channel autopilot ●

a. parallel processing, mutual monitoring and output comparison
b. mechanical voting, torque limiting and authority limiting
c. active redundancy, crosstalk and autopilot synchronisation

LEVEL 3 — APPLY AND REASON

219. An autopilot that detects its own failure and disengages itself is operated on its own. Which behaviour does the system then show? ●

a. fail passive behaviour
b. fail operational behaviour
c. fail safe behaviour

220. A manufacturer limits the authority of its autopilot servos, but the aircraft has no manual control path strong enough to overrule the autopilot. Why is that not acceptable? ●

a. because the autopilot could then not recognize its own hard over
b. because the pilot could then not over-power the autopilot
c. because the servos would then not reach their full travel range

221. A fail passive system must be able to disengage itself. How does it recognize that it has a failure? ●

a. by monitoring the hydraulic pressure at the actuator
b. by comparing the two autopilot computers with each other
c. by comparing the surface position with the commanded position

222. During dual operation the two autopilot actuators try to deflect one surface in opposite directions. What does the surface do?

a. it runs to its full deflection
b. it stays in its position
c. it follows the faster of the two actuators

223. Two fail safe autopilots drive one surface in dual operation. What does the feel system contribute?

a. the neutral reference for the deflection limiter
b. the torque limit applied at the actuator clutch
c. the third signal the mid-value voter needs

224. In which respect does a dual channel fail passive autopilot differ from two parallel fail safe autopilots ●

a. it drives three actuators and needs a mid-value voter
b. it drives one actuator only and needs no mechanical voting
c. it drives two actuators and needs a torque limiter

Answers — Fail safe and fail passive systems

198. a That wording is the definition of the flight control system. (L1, PDF p. 288)

199. a The definition limits the term landing system to equipment carried on board. (L1, PDF p. 288)

200. b A hard over of a control surface is named as the most important feature of the fail safe failure case. (L1, PDF p. 290)

201. b FAR 25 allows 3 seconds for the takeover enroute. (L1, PDF p. 290)

202. b Two fail safe autopilots working in parallel are described as active redundancy. (L1, PDF p. 292)

203. b Fail passive behaviour can come from two independent channels inside one autopilot. (L1, PDF p. 294)

204. a A detected failure disengages the autopilot. (L1, PDF p. 294)

205. b The equipment named in the definition is sensors, instruments and power supplies. (L2, PDF p. 288)

206. c The architectures exist to produce the fail passive or fail operational modes that CAT III operation accepts. (L2, PDF p. 288)

207. a Dual operation of fail passive single systems yields fail operational behaviour. (L2, PDF p. 288)

208. a Suppression is impossible because the system has no way of recognizing the failure. (L2, PDF p. 290)

209. c Reducing servo authority is the means given for limiting the failure effect. (L2, PDF p. 290)

210. a A manual flight control system able to over-power the autopilot is the condition for allowing authority limitation. (L2, PDF p. 290)

211. c A fail passive failure leaves the trim state unchanged and causes no considerable deviation from flight path or attitude. (L2, PDF p. 292)

212. b The printed definition says an automatic landing can no longer be performed. (L2, PDF p. 292)

213. c On failure of a fail passive system the pilot has to take over control. (L2, PDF p. 292)

214. b Independence comes from everything being separate: supplies, sensors, computers, actuators and cable ducts. (L2, PDF p. 292)

215. a Force conflicts are attributed to tolerances, pressure differences and poor adjustment. (L2, PDF p. 292)

216. b The channels differ in hardware and in the programming language of the identical programs. (L2, PDF p. 294)

217. b The outputs are compared for validity, and only then released to the actuator. (L2, PDF p. 294)

218. a Those three together are what makes the dual channel autopilot fail passive. (L2, PDF p. 294)

219. a A single system able to detect and remove its own failure gives fail passive behaviour in single operation. (L3, PDF p. 288)

220. b Authority limitation relies on the pilot being able to apply strong inputs and over-power the autopilot manually. (L3, PDF p. 290)

221. c Recognition rests on the electrical comparison of actual surface position with commanded actuator position. (L3, PDF p. 292)

222. b Opposing actuator commands are voted out, so the surface holds its position. (L3, PDF p. 292)

223. c Mid-value voting needs three inputs, and with two autopilots the feel system provides the missing one. (L3, PDF p. 292)

224. b Parallel fail safe autopilots need mechanical voting against force conflicts, while the dual channel autopilot drives a single actuator and does not. (L3, PDF p. 294)

Fail operational and hybrid landing systems

PDF pages 296–302 · 15 questions

LEVEL 1 — RECALL

225. Of which two parts does a hybrid landing system consist ●

a. a fail safe landing system and a mechanical voter
b. a fail operational landing system and a second autopilot
c. a fail passive landing system and a guidance system

226. Which systems are usually used as the supporting system of a hybrid landing system ●

a. yaw damper systems
b. head up guidance systems
c. radio altimeter systems

LEVEL 2 — UNDERSTAND AND EXPLAIN

227. Why does mechanical voting have to be applied in triple operation

a. to prevent force conflicts
b. to detect a hard over of a surface
c. to limit the authority of the servos

228. Two autopilots run in parallel and each one disengages itself on its own failure. Which kind of redundancy does that architecture show? ●

a. standby redundancy
b. passive redundancy
c. active redundancy

229. The main system of a hybrid landing system fails on the approach. What does the supporting guidance system then allow? ●

a. automatic continuation of approach and landing
b. an automatic go around at the alert height
c. manual continuation of approach and landing

230. Of which units does the hybrid built from fail safe autopilots consist ●

a. one fail safe autopilot and two flight director systems
b. three fail safe autopilots and one head up display
c. two fail safe autopilots and one flight director system

231. Of which units does the hybrid system built around a fail passive autopilot consist ●

a. one fail passive autopilot and two head up displays
b. one fail passive autopilot and one flight director system
c. two fail passive autopilots and one flight director system

LEVEL 3 — APPLY AND REASON

232. A single autopilot fails below the alert height in a fail operational system. What can the crew still expect of the automatic system? ●

a. the system disengages and the approach has to be abandoned
b. approach, flare, landing and roll out are completed automatically
c. the approach continues but the landing has to be flown manually

233. A fail operational system has already suffered one failure. What would a second failure mean now? ●

- **a.** a hard over may occur, as the system is now fail safe
- **b.** the landing is still completed, as the system stays fail operational
- **c.** the pilot must take over, as the system is now fail passive

234. In a triple operation system one autopilot output drifts away from the other two. What does the mid-value voter do to the surface?

- **a.** the surface follows the middle actuator and the odd output is lost
- **b.** the surface follows the odd output until that autopilot is off
- **c.** the surface takes the average, so the odd output shifts it

235. Two fail passive autopilots work in parallel on the same control train. Why can no force conflict arise there? ●

- **a.** because a mid-value voter selects the middle of the signals
- **b.** because only one actuator drives the surface at a time
- **c.** because a torque limiter is fitted in each actuator clutch

236. One of two parallel fail passive autopilots detects its own failure during an automatic landing. What happens next? ●

- **a.** both of them disengage and the pilot takes over
- **b.** the faulty one is voted out by the mechanical voter
- **c.** it disengages itself and the other one keeps control

237. Which behaviour does the autopilot system of a hybrid built from two fail safe autopilots show ●

- **a.** fail safe, augmented by an auto-thrust system
- **b.** fail passive, augmented by a flight director
- **c.** fail operational, augmented by a flight director

238. The autopilot of a hybrid landing installation fails on short final. How can the landing still be completed? ●

- **a.** automatically, by the second autopilot taking over
- **b.** manually, following the supporting guidance system
- **c.** automatically, after the voter has selected a channel

239. Why does a hybrid landing system count as fail operational although its automatic landing system is only fail passive ●

- **a.** the automatic landing system becomes fail operational below the alert height
- **b.** the two autopilots of the system are voted mechanically
- **c.** the guidance system allows the landing to be continued manually

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Answers — Fail operational and hybrid landing systems

225. c The main system is a fail passive automatic landing system and it is supported by an independent guidance system. (L1, PDF p. 300)

226. b Head up guidance systems are the usual supporting systems. (L1, PDF p. 300)

227. a Voting is applied for the same reason as in dual operation: to prevent force conflicts. (L2, PDF p. 296)

228. b This architecture is described as passive redundancy. (L2, PDF p. 298)

229. c The supporting system lets the crew carry on manually after the main system has failed. (L2, PDF p. 300)

230. c It is built from two fail safe autopilots plus at least one flight director system. (L2, PDF p. 300)

231. b This architecture uses a single fail passive autopilot with at least one flight director system. (L2, PDF p. 302)

232. b Fail operational means the whole sequence down to roll out is still flown automatically. (L3, PDF p. 296)

233. c Once fail passive, a further failure means no automatic landing and the pilot has to take over. (L3, PDF p. 296)

234. a Mid-value voting passes only the middle signal, so a single deviating output is suppressed. (L3, PDF p. 296)

235. b At any instance only one actuator is driving, so nothing can work against it. (L3, PDF p. 298)

236. c The failed autopilot takes itself out and the remaining one continues to control the aircraft. (L3, PDF p. 298)

237. b Two fail safe autopilots make the autopilot system fail passive, and the flight director augments it. (L3, PDF p. 300)

238. b The hybrid system's reserve is manual flight on the supporting guidance system, not a second automatic system. (L3, PDF p. 300)

239. c The overall capability is kept by the independent guidance system letting the crew finish the landing by hand. (L3, PDF p. 300)

Landing capabilities, autoland warning and equipment

PDF pages 304–310 · 14 questions

LEVEL 1 — RECALL

240. Where is the autoland warning displayed ●

a. on the overhead panel above the pilots
b. on the engine display of the lower screen
c. in the glare shield, in the pilot's view

241. In which colour is the autoland warning displayed ●

a. red
b. white
c. amber

242. Where are components influencing CAT II and CAT III approach capability specially marked ●

a. in column 3 of the minimum equipment list
b. in column 1 of the operations manual part B
c. in chapter 22 of the aircraft maintenance manual

LEVEL 2 — UNDERSTAND AND EXPLAIN

243. The capability logic has checked the serviceability of all required systems. Which supplies does it count as well?

a. the independent pneumatic and electrical power supplies
b. the independent hydraulic and pneumatic power supplies
c. the independent hydraulic and electrical power supplies

244. What does the degree of redundancy determine ●

a. the failure mode of the system
b. the decision height of the approach
c. the alert height of the approach

245. One fail passive autopilot or flight director is operational and the redundancy is limited. Which capability is reached? ●

a. CAT 3 SINGLE
b. CAT 1
c. CAT 2

246. The autoland warning comes on during an automatic approach. Which reaction is the normal one? ●

a. the initiation of a go around
b. the disengagement of one autopilot
c. a continuation of the approach to CAT I minima

247. The autoland warning comes on and the runway is clearly in sight. What is possible besides a go around? ●

a. a manual landing
b. an automatic landing
c. a second automatic approach

248. Above which difference of radio height between two radio altimeters does the autoland warning come on

- **a.** 50 ft
- **b.** 100 ft
- **c.** 10 ft

LEVEL 3 — APPLY AND REASON

249. Required ground equipment fails while the aircraft is still above the alert height and the capability changes. What has to be done? ●

- **a.** the autopilot has to be disengaged and the approach flown manually
- **b.** the approach may be continued to CAT I minima
- **c.** a go around has to be initiated

250. A single failure occurs below the alert height and degrades the failure mode from fail operational to fail passive. What happens to the displayed capability? ●

- **a.** it is no longer changed
- **b.** it is replaced by a red warning
- **c.** it drops by one category

251. A single failure below the alert height degrades the failure mode, yet the capability stays as it was. Why is it kept? ●

- **a.** because the ground equipment can no longer be monitored
- **b.** because the autopilot can no longer be disengaged
- **c.** because the autoflight system still performs its duty

252. One fail passive autopilot and one auto-thrust system are operational. Which capability is displayed? ●

- **a.** CAT 2
- **b.** CAT 3 DUAL
- **c.** CAT 3 SINGLE

253. Two fail passive autopilots and one auto-thrust system are operational. Which capability is displayed and which failure mode does the system have? ●

- **a.** CAT 3 SINGLE, and the system is fail passive
- **b.** CAT 2, and the system is fail operational
- **c.** CAT 3 DUAL, and the system is fail operational

Answers — Landing capabilities, autoland warning and equipment

240. c It sits in the glare shield, inside the pilot's field of view. (L1, PDF p. 306)

241. a The autoland warning is a red warning. (L1, PDF p. 306)

242. a The MEL marks them in its column 3. (L1, PDF p. 310)

243. c The number of independent hydraulic and electrical power supplies is counted as well. (L2, PDF p. 304)

244. a Redundancy decides whether the system is fail safe, fail passive or fail operational. (L2, PDF p. 304)

245. b With one fail passive autopilot or flight director and limited redundancy only CAT I is available. (L2, PDF p. 304)

246. a A go around is the normal response to the autoland warning. (L2, PDF p. 306)

247. a With visual conditions permitting, the landing may be completed manually. (L2, PDF p. 306)

248. c The warning condition is a radio height difference of more than 10 ft. (L2, PDF p. 306)

249. c A capability change above the alert height requires a go around. (L3, PDF p. 304)

250. a Below the alert height the capability is frozen even though the failure mode degrades. (L3, PDF p. 304)

251. c The capability stays because the system is still doing its job after the single failure. (L3, PDF p. 304)

252. c That combination gives CAT III A, displayed as CAT 3 SINGLE. (L3, PDF p. 304)

253. c Two fail passive autopilots make the total system fail operational and give CAT 3 DUAL for CAT III B. (L3, PDF p. 304)

Helicopter stability augmentation

PDF pages 312–320 · 40 questions

LEVEL 1 — RECALL

254. What does the collective pitch lever control ●

a. the velocity vector
b. the main rotor lift
c. the tail rotor thrust

255. What does the cyclic pitch stick control ●

a. the velocity vector
b. the main rotor lift
c. the tail rotor thrust

256. The cyclic stick is moved forward. Which movement of the helicopter results? ●

a. a nose down movement
b. a tail down movement
c. a left sideward movement

257. Through which servo do the pedals act ●

a. the LH servo
b. the yaw servo
c. the front servo

258. What is a twist grip throttle ●

a. a friction lock fitted at the pedal unit
b. a rotary rpm controller at the collective pitch lever
c. a rotary trim controller at the cyclic pitch stick

259. On which helicopters is a twist grip throttle found ●

a. on those powered by a turbine engine
b. on those powered by a piston engine
c. on those with a counter clockwise rotor

260. Which stability does a helicopter have around its longitudinal, lateral and vertical axes ●

a. positive stability around the longitudinal axis only
b. no positive stability around any of them
c. positive stability around the vertical axis only

261. Which sensors does a helicopter stabilization system use ●

a. rate gyros
b. accelerometers
c. pressure transducers

LEVEL 2 — UNDERSTAND AND EXPLAIN

262. The collective pitch lever is raised. How is the pitch angle of the main rotor blades changed? ●

a. on half of the blades at once and by the same amount
b. on all blades at once and by the same amount
c. on each blade in turn and by a different amount

263. How does the cyclic stick produce a variation of the rotor blade pitch angle ●

a. by changing the pitch of all the blades equally
b. by changing the thrust produced by the tail rotor
c. by tilting the main rotor plane around the X or Y axis

264. Only the front servo is driven by a control input. Around which axis is the rotor plane tilted? ●

a. the lateral axis, Y-Y'
b. the vertical axis, Z-Z'
c. the longitudinal axis, X-X'

265. Which servos tilt the rotor plane around the longitudinal axis ●

a. the front and LH servos
b. the front and yaw servos
c. the LH and RH servos

266. What is the up and down movement of the swash plate called ●

a. collective pitch variation
b. lateral pitch variation
c. cyclic pitch variation

267. The swash plate is moved up from its initial position. Which quantity at the rotor changes? ●

a. the length of the blade lever rods
b. the rotational speed of the main rotor
c. the angle of attack of the rotor blades

268. What determines the axis about which the swash plate is tilted ●

a. the position of the two pedal units
b. the amount of collective pitch applied
c. the movement direction of the cyclic stick

269. The swash plate is to be tilted about the lateral axis. Which servo is energized on its own? ●

a. the yaw servo
b. the LH servo
c. the front servo

270. A sideward movement is commanded. How are the LH and RH servos driven for it? ●

a. one after the other, with signals of different magnitude
b. concurrently, with equal signals in opposite directions
c. concurrently, with equal signals in the same direction

271. Which movement of the swash plate does a diagonal cyclic stick movement produce ●

a. a combined tilt about both of the axes
b. an up and down movement without any tilt
c. a tilt about the lateral axis only

272. The pilot presses one of the pedals. What is changed at the tail rotor? ●

a. the diameter of the tail rotor disc
b. the pitch angle of the tail rotor blades
c. the rotational speed of the tail rotor

273. Around which axis does the tail rotor allow the helicopter to be controlled ●

a. the vertical axis
b. the lateral axis
c. the longitudinal axis

274. What gives an aeroplane the stability a helicopter lacks ●

a. the rate gyros feeding its stabilization computer
b. the attenuation signals sent to its control servos
c. the dihedral and sweep of the wing and the stabilizers

275. Why have stabilization systems been developed for helicopters ●

a. to let the pilot concentrate on navigation and supervision
b. to allow the main rotor to be driven at a lower speed
c. to raise the maximum forward speed of the helicopter

276. The SCAS has received a rate signal from one of the gyros. What does it drive with its attenuation signal? ●

a. the trim actuators of the cyclic stick
b. the servo valves of the control servos
c. the main servo input levers directly

277. Which stability does the helicopter show once the stabilization system is working ●

a. neutral stability
b. positive stability
c. negative stability

278. What holds the cable system of the main rotor controls in position ●

a. the control servos
b. the motion transducers
c. the trim actuators

279. The control servos of a stabilized helicopter are driven by the SCAS computer. What do they move? ●

a. the swash plate directly
b. the main servo input levers
c. the trim control panel switches

280. The pilot returns the cyclic stick to its neutral position. What does the output of the motion transducer do? ●

a. it becomes zero
b. it reaches its maximum
c. it reverses its sign

281. From which sources is the stored reference value obtained ●

a. from the rate gyros, via the stability augmentation computer
b. from the attitude and heading gyros, via the synchronizer
c. from the motion transducers, via the trim actuator units

282. To which fixed wing function is the helicopter's stored attitude reference mode comparable ●

a. to the autothrottle
b. to control wheel steering
c. to the yaw damper

283. What is compared in the flight path control of a helicopter

- **a.** the planned path and altitude with the momentary path parameters
- **b.** the commanded path with the signals of the three rate gyros
- **c.** the trim signals with the outputs of the motion transducers

284. A flight path error has been worked out on a helicopter flying an upper mode. Where is it superimposed on the stabilization signals? ●

- **a.** in the trim control panel
- **b.** in the SCAS computer
- **c.** in the synchronizer control unit

285. Through which item does each trim signal reach its trim actuator ●

- **a.** the synchronizer control unit
- **b.** the respective motion transducer
- **c.** the stability augmentation computer

LEVEL 3 — APPLY AND REASON

286. The cyclic stick is moved to the left-hand side. Which servo stays unenergized and which movement results? ●

- **a.** the front servo, and the helicopter moves left sideward
- **b.** the front servo, and the helicopter moves nose down
- **c.** the LH servo, and the helicopter moves left sideward

287. The pedal position is changed in flight. Which quantity at the tail rotor changes as a result of the blade pitch change? ●

- **a.** the thrust of the tail rotor
- **b.** the torque of the main rotor
- **c.** the speed of the tail rotor

288. With speed already established the cyclic stick is moved back. Which pair of reactions does that give?

- **a.** a descent like an airplane, with speed being lost
- **b.** a climb like an airplane, with speed being gained
- **c.** a climb like an airplane, with speed being lost

289. With speed established the left pedal is applied. Which pair of reactions results?

- **a.** a yaw to the left, with a little speed increase
- **b.** a yaw to the right, with a little speed drop
- **c.** a yaw to the left, with a little speed drop

290. What follows from the helicopter having no positive stability ●

- **a.** its rotor speed has to be controlled permanently by the pilot
- **b.** its attitude has to be controlled permanently by the pilot
- **c.** its trim actuators have to run permanently in flight

291. How does the pilot get feedback of the attitude change at his controls ●

- **a.** through the rate gyros, driven by the SCAS computer
- **b.** through the trim actuators, driven by the control panel
- **c.** through the cable system, driven by the main servos

292. The cyclic stick is returned to neutral after a manoeuvre. What does the SCAS computer do with the attitude reached at that moment? ●

a. it stores it as the reference value to be maintained
b. it compares it with the planned flight path of the mission
c. it passes it to the trim actuators as a trim command

293. A pilot input and the stabilization signals are present at the same time. How does the SCAS computer handle them? ●

a. it blocks the stabilization signals while the pilot inputs
b. it overlays the attitude reference and stabilization signals
c. it passes only the larger of the two signals to the servos

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Answers — Helicopter stability augmentation

254. b The collective pitch lever is the lift control. (L1, PDF p. 312)

255. a The cyclic stick controls the velocity vector. (L1, PDF p. 312)

256. a Forward cyclic gives a nose down movement. (L1, PDF p. 314)

257. b Pedal inputs pass through the yaw servo. (L1, PDF p. 316)

258. b It is the rotary rpm control sitting on the collective pitch lever. (L1, PDF p. 318)

259. b The twist grip throttle belongs to piston engined helicopters. (L1, PDF p. 318)

260. b The helicopter has no positive stability about any of the three axes. (L1, PDF p. 320)

261. a Rotation rates about pitch, roll and yaw are sensed by rate gyros. (L1, PDF p. 320)

262. b Collective input acts on every blade simultaneously and equally. (L2, PDF p. 312)

263. c The cyclic tilts the rotor plane about the X or Y axis. (L2, PDF p. 312)

264. a The front servo tilts the rotor plane about the lateral Y-Y' axis. (L2, PDF p. 312)

265. c The LH and RH servos work the longitudinal X-X' axis. (L2, PDF p. 312)

266. a That up and down movement is the collective pitch variation. (L2, PDF p. 314)

267. c Moving the swash plate up and down alters the blades' angle of attack. (L2, PDF p. 314)

268. c The tilt axis follows the direction in which the cyclic stick is moved. (L2, PDF p. 314)

269. c Only the front servo is energized for a tilt about the lateral axis. (L2, PDF p. 314)

270. b They get equal signals at the same time, but run opposite to each other. (L2, PDF p. 314)

271. a A diagonal input tilts the swash plate proportionally about both axes at once. (L2, PDF p. 314)

272. b The pedals set tail rotor blade pitch. (L2, PDF p. 316)

273. a Tail rotor thrust gives control about the vertical axis, that is yaw control. (L2, PDF p. 316)

274. c Wing dihedral and sweep together with the stabilizers give the fixed wing aircraft its stability. (L2, PDF p. 320)

275. a They relieve the pilot of the constant control task so he can attend to navigation and supervision. (L2, PDF p. 320)

276. b The attenuation signals go to the servo valves of the control servos. (L2, PDF p. 320)

277. a With the stabilization system active the helicopter exhibits neutral stability. (L2, PDF p. 320)

278. a The control servos hold that cable system in position. (L2, PDF p. 320)

279. b The control servos act on the main servo input levers. (L2, PDF p. 320)

280. a At neutral the motion transducer output is zero. (L2, PDF p. 320)

281. b It comes from the pitch-roll attitude gyro and the heading gyro through the synchronizer control unit. (L2, PDF p. 320)

282. b Its behaviour is comparable to control wheel steering. (L2, PDF p. 320)

283. a Planned path and altitude are compared against the momentary path parameters. (L2, PDF p. 320)

284. b The superposition happens inside the SCAS computer. (L2, PDF p. 320)

285. a Trim signals pass through the synchronizer control unit on their way to the trim actuator. (L2, PDF p. 320)

286. a A sideward input leaves the front servo unenergized and moves the helicopter sideward to the left. (L3, PDF p. 314)

287. a Changing tail rotor blade pitch changes the tail rotor thrust. (L3, PDF p. 316)

288. c With speed on, a rearward cyclic input gives an airplane like climb and the speed falls off. (L3, PDF p. 319)

289. c Left pedal with speed on yaws the helicopter left and costs a little speed. (L3, PDF p. 319)

290. b Without inherent stability the pilot has to control the attitude all the time. (L3, PDF p. 320)

291. c The main servos feed the movement back to the pilot controls through the cable system. (L3, PDF p. 320)

292. a The momentary attitude becomes the new reference the stabilization system holds. (L3, PDF p. 320)

293. b The two are overlaid on each other inside the computer rather than one replacing the other. (L3, PDF p. 320)